Recap: Series - $\sum_{k=1}^{\infty} a_k$ - Formal expressions.

Sequence of partial sums - (S_n) - $S_n = a_1 + \dots + a_n$.

Building tests on (a_k) that tell us something about (S_n).

- Divergence Test - If $\lim(a_n) \neq 0$ or does not exist then the series $\sum_{k=1}^{\infty} a_k$ diverges.

- Cauchy Condensation Test - $\sum_{k=0}^{\infty} 2^k a_{2^k}$.

We have seen that $\sum_{k=1}^{\infty} \frac{1}{k^2}$ is convergent yet the series $\sum_{k=1}^{\infty} \frac{1}{k}$ is divergent. We would perhaps expect that $\sum_{k=1}^{\infty} \frac{1}{k^p}$, where $p \in \mathbb{N}$, $p \geq 2$, is convergent. What about $\sum_{k=1}^{\infty} \frac{1}{k^{1.5}}$? Is this convergent?.

Theorem 4.6: (p-series Test)

The series

$$\sum_{k=1}^{\infty} \frac{1}{k^p}$$

is convergent iff $p > 1$.

Proof: By the Cauchy condensation test it follows that $\sum_{k=1}^{\infty} \frac{1}{k^p}$ is convergent iff

$$\sum_{k=0}^{\infty} 2^k \left(\frac{1}{2^k}\right)^p = \sum_{k=0}^{\infty} \left(\frac{1}{2^{p-1}}\right)^k$$

is also convergent. This series is geometric and we know this is convergent iff $\left|\frac{1}{2^{p-1}}\right| < 1$. Since $2^x > 1$ for all $x > 0$ and $2^x \leq 1 \ \forall \ x \leq 0$ it follows that $\left|\frac{1}{2^{p-1}}\right| < 1$ iff $p > 1$. ☐

By comparing general series to series of the form $\sum_{k=1}^{\infty} \frac{1}{k^p}$ we can sometimes determine the nature of the general series.

Theorem 4.7 ; Comparison Test

Let (a_k) and (b_k) be sequences such that $0 \leq a_k \leq b_k$ for all $k \in \mathbb{N}$.

i) If $\sum_{k=1}^{\infty} b_k$ converges then $\sum_{k=1}^{\infty} a_k$ converges.

ii) If $\sum_{k=1}^{\infty} a_k$ diverges then $\sum_{k=1}^{\infty} b_k$ diverge.

Proof: (i) Suppose the $\sum_{k=1}^{\infty} b_k$ converges. By the Cauchy Criterion on Series it follows that for all $\varepsilon > 0$, there exists $N \in \mathbb{N}$ s.t

$$|b_{m+1} + b_{m+2} + \dots + b_n| < \varepsilon$$

for all $n > m > N$. Since $0 \leq a_k \leq b_k$ $\forall k \in \mathbb{N}$ it follows that

$$|a_{m+1} + a_{m+2} + \dots + a_n| < \varepsilon$$

$\forall n > m > N$. But by the Cauchy criterion for series it follows that $\sum_{k=1}^{\infty} a_k$ is convergent. ☐

Ex: (Comparison Test)

(i) Consider the series $\sum_{k=1}^{\infty} \frac{k+4}{k^3+1}$.

Clearly $\frac{k+4}{k^3+1} > 0$ for all $k \in \mathbb{N}$. Additionall,

$$\frac{k+4}{k^3+1} \leq \frac{k+4k}{k^3+1} = \frac{5k}{k^3+1}$$

$$\leq \frac{5k}{k^3}$$

$$= \frac{5}{k^2}.$$

Since $\sum_{k=1}^{\infty} \frac{5}{k^2}$ is convergent by the p-series test and the algebraic limit theorem for series, it follows by the comparison test that $\sum_{k=1}^{\infty} \frac{k+4}{k^3+1}$ is convergent.

iii) Consider the series

$$\sum_{k=1}^{\infty} \frac{(k+1)\sin^2(k)}{k^3+3k+1}.$$

Clearly $\frac{(k+1)\sin^2(k)}{k^3+3k+1} \geq 0$ for all $k \in \mathbb{N}$. Now,

$$\frac{(k+1)\sin^2(k)}{k^3+3k+1} \leq \frac{k+1}{k^3+3k+1} \leq \frac{k+1}{k^3}$$

$$\leq \frac{k+k}{k^3} = \frac{2}{k^2}.$$

Hence, by the p-series test and algebra of convergent series it follows that $\sum_{k=1}^{\infty} \frac{2}{k^2}$ is convergent. Thus, by the comparison test $\sum_{k=1}^{\infty} \frac{(k+1)\sin^2(k)}{k^3+3k+1}$ is convergent.

We have seen in the early lectures the series $\sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k}$.

Theorem 4.9: Alternating Series Test

Let (a_k) be a sequence such that

i) (a_k) is monotone decreasing and

ii) $\lim(a_k) = 0$.

Then, the alternating series $\sum_{k=1}^{\infty} (-1)^k a_k$ converges.

Proof: Omitted.